

Anticipating demand, not only carbon: receding-horizon scheduling of deferrable computing workloads across data centres with calibrated carbon-intensity forecasts

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Abstract

Shifting deferrable jobs to cleaner hours and locations cuts data-centre emissions, but online schedulers must act before future grid conditions and jobs are known. This paper formulates multi-site, deadline-constrained scheduling of bag-of-tasks jobs as a minimum-cost flow problem and proposes CARMA (Carbon-aware Anticipatory Receding-horizon Multi-site Allocation). CARMA adds “ghost jobs” for expected arrivals to each hourly plan and uses a 1–24-hour carbon-intensity forecaster with adaptive conformal calibration. Three guarantees are proved. First, uncertainty about demand forces any online algorithm, even one with perfect carbon forecasts, to a competitive ratio of order $\sqrt{\theta}$, where θ is the ratio of the highest to the lowest unit emission; myopic planning is worse, at $(1 + \theta)/2$. Second, given backstop capacity, CARMA never misses a deadline and is θ -competitive under any predictions, is optimal under exact predictions, and degrades linearly with prediction error. Third, dual prices of the pooled problem yield a carbon attribution among sites that lies in the core, so no coalition gains by leaving. On three years of regional grid data for Great Britain, with 52 out-of-sample weeks in 2024, CARMA reduces emissions by 36–68% relative to carbon-agnostic operation when migration is allowed. It comes within 5.8–8.7% of the clairvoyant optimum, against 17.6–21.5% for myopic control, and lowers late work. Anticipating demand is worth more than perfect carbon foresight, and CARMA degrades gracefully under injected errors. Location-based carbon accounting violates coalitional stability in every test week, whereas the dual attribution

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satisfies it exactly in hindsight and in over 99% of checks online.

Keywords: Carbon-aware computing, Data centre scheduling, Model predictive control, Minimum-cost flow, Conformal prediction, Carbon intensity forecasting

1. Introduction

Data centres are among the fastest-growing electricity consumers. Efficiency gains kept their energy use nearly flat through the 2010s even as computing output grew several-fold [1, 2]. That buffer is now thinning, and information and communication technology accounts for an estimated 2.1–3.9% of global greenhouse-gas emissions [3]. Large machine-learning workloads have drawn particular attention [4, 5, 6]. The carbon intensity of grid electricity varies by a factor of ten or more across regions and hours, driven by the output of wind, solar and dispatchable plant [7]. Operators can therefore cut emissions without new hardware by moving flexible work to cleaner places and times. Such *carbon-aware computing* [8, 9] is an operations problem: capacity is finite, deadlines bind, and both the grid and the job stream are uncertain.

Batch jobs such as model training, analytics, rendering and backups are a natural target, because they tolerate some delay. Previous work has shown the potential of temporal shifting (running later) [10, 11, 12] and of spatial shifting (running elsewhere) [13, 14]. It has also shown their limits. Sukprasert et al. [15] found that practical reductions from spatio-temporal shifting fall well short of idealised bounds. Most methods are evaluated with a scheduler that reacts to the jobs it has already seen, using point forecasts of carbon intensity. Three questions have received less attention. First, how much is lost because an online scheduler does not know which jobs will arrive next, and can any scheduler guarantee good performance in the worst case? Second, does it pay to price the uncertainty of carbon-intensity forecasts into scheduling decisions? Third, when several sites pool their capacity, how should the resulting emissions be attributed so that every participant is better off than on its own?

This paper answers these questions with a new algorithm, provable guarantees and a forecasting model, evaluated on three years of real regional grid data for Great Britain. It makes five contributions.

- 31 1. **Formulation.** We state multi-site, deadline-constrained, width-limited
32 scheduling of deferrable bag-of-tasks jobs, including the emissions of
33 data transfer, as a minimum-cost flow problem (Proposition 1). Of-
34 fline clairvoyant optima and every online planning step can therefore
35 be computed exactly and quickly by network simplex. We also prove
36 that myopic receding-horizon scheduling can be arbitrarily worse than
37 the clairvoyant optimum even with perfect carbon forecasts (Proposi-
38 tion 2).
- 39 2. **Algorithm.** We propose CARMA (Carbon-aware Anticipatory Receding-
40 horizon Multi-site Allocation). At every hour, CARMA adds *ghost jobs*
41 to its planning network. These represent expected arrivals, learned
42 from the operator’s history, so the planner reserves scarce low-carbon
43 capacity for work that has not yet arrived. The planning problem stays
44 a single min-cost flow and decisions take tens of milliseconds.
- 45 3. **Guarantees.** We prove that uncertainty about demand alone forces
46 every online algorithm, even one with perfect carbon forecasts, to
47 a competitive ratio of at least about $\sqrt{\theta}/2$, where θ is the ratio of
48 the highest to the lowest unit emission (Theorem 1). We show that
49 CARMA is robust, consistent and smooth: given a backstop resource,
50 it never misses a deadline and is θ -competitive whatever its predictions,
51 it is optimal when they are exact, and its excess emissions grow at most
52 linearly in the prediction errors (Theorem 2). We also show that the
53 dual prices of the pooled scheduling network give a carbon attribution
54 among sites that lies in the core of the pooling game, so no group of
55 sites gains by leaving (Theorem 3).
- 56 4. **Forecasting model.** We develop a probabilistic 1–24-hour carbon-
57 intensity forecaster. It consists of direct gradient-boosted models with
58 horizon-wise normalised conformal calibration and adaptive conformal
59 inference, which holds coverage under the strong decarbonisation drift
60 seen in 2024. Its calibrated quantiles let us test risk-adjusted carbon
61 pricing inside the scheduler.
- 62 5. **Evidence.** Across 52 out-of-sample weeks, three load levels and two
63 operating regimes, we compare CARMA with five implementable on-
64 line baselines, clairvoyant and ablated variants, and the clairvoyant
65 oracle. The results separate the value of anticipating demand from the
66 value of better carbon forecasts, show when each matters, confirm the
67 graceful degradation predicted by the theory, and show that common
68 location-based carbon accounting violates coalitional stability in every

69 test week, whereas the dual attribution satisfies it exactly for the hind-
70 sight benchmark and in over 99% of the checks for CARMA’s realised
71 emissions.

72 The results carry an operational message. Once spatial shifting is allowed,
73 the dominant loss comes from demand myopia, not from forecast error. An-
74 ticipating demand cuts the gap to the clairvoyant optimum by a factor of
75 2.2–3.0. By contrast, perfect carbon foresight or risk-averse carbon pricing
76 adds little once demand is anticipated. Without migration, temporal shift-
77 ing alone achieves much smaller reductions. The forecast-driven planners
78 are then within 4–6% of the optimum, and the remaining gap is due almost
79 entirely to forecast error. The rest of the paper is organised as follows. Sec-
80 tion 2 reviews related work. Section 3 presents the model, the forecaster,
81 CARMA and the experimental design. Section 4 establishes the theoretical
82 guarantees, with proofs in Appendix A. Section 5 reports results, Section 6
83 discusses implications and limitations, and Section 7 concludes.

84 2. Related work

85 *Carbon-aware workload shifting.* Early green data-centre schedulers matched
86 deferrable jobs to on-site solar production, for example GreenSlot [16] and
87 GreenHadoop [11]. Other work exploited electricity-price and carbon differ-
88 ences across sites through geographical load balancing [17, 13] and dynamic
89 right-sizing [18]. With public carbon-intensity feeds, attention moved to grid
90 emissions. Wiesner et al. [10] quantified the reductions from temporal shift-
91 ing in several regions, including Great Britain, and proposed forecast-based
92 lowest-window scheduling. Google’s carbon-intelligent compute system sets
93 next-day virtual capacity curves from carbon and demand forecasts [8]. Car-
94 bonScaler varies the allocation of elastic jobs with carbon intensity [12]. Eco-
95 visor exposes the energy system to applications [19]. Carbon Explorer weighs
96 scheduling against investment in renewables and storage [20]. Load migra-
97 tion between data centres can absorb curtailed renewable generation [14],
98 and data-centre demand response has been surveyed in [21]. Sukprasert et
99 al. [15] bound the potential of spatio-temporal shifting across 123 regions and
100 find that simple policies capture most of the achievable benefit. Our study
101 complements theirs by identifying *which* information limits online schedulers
102 when capacity at clean sites is scarce.

103 *Online algorithms and guarantees.* Worst-case analysis of carbon-aware schedul-
 104 ing builds on online search and one-way trading [22], for which threshold
 105 policies achieve optimal competitive ratios. Lechowicz et al. derived op-
 106 timal double-threshold algorithms for the online pause-and-resume prob-
 107 lem [23]. They also derived learning-augmented algorithms with an opti-
 108 mal consistency–robustness trade-off for spatiotemporal allocation of a sin-
 109 gle deadline-constrained workload over a network of data centres [24]. These
 110 works model uncertainty about future carbon *prices* for one job. We study
 111 many jobs that compete for finite low-carbon capacity, and we show that un-
 112 certainty about *demand* creates a distinct lower bound that carbon forecasts
 113 cannot remove. Our guarantees follow the consistency–robustness paradigm
 114 of algorithms with predictions [25]. CARMA’s re-solving of a determinis-
 115 tic plan with expected future demand is related to re-solving heuristics in
 116 network revenue management [26]. Our fairness result draws on cooperative
 117 game theory: linear production games and flow games have non-empty cores
 118 that can be computed from dual prices [27, 28].

119 *Energy-aware scheduling in operations.* Energy-efficient scheduling is a ma-
 120 ture topic in manufacturing operations [29], and green scheduling models
 121 have appeared in this journal, for example for parallel machines with job
 122 splitting [30] and for cloud resource allocation [31]. Most such models are de-
 123 terministic and solved offline, often with metaheuristics. We instead exploit
 124 exact network-flow structure [32, 33] and embed it in an online lookahead
 125 policy. In the taxonomy of Powell [34], CARMA is a deterministic lookahead
 126 policy with a modified (anticipatory) model, in the spirit of model predictive
 127 control [35, 36].

128 *Carbon-intensity forecasting.* Forecasts of grid carbon intensity use statistical
 129 decomposition [37], machine learning with many exogenous drivers [38], day-
 130 ahead regression for building demand response [39], and hierarchical models
 131 of the generation mix, as in CarbonCast [40]. For Great Britain, NESO
 132 publishes regional forecasts through its Carbon Intensity API [41]. These
 133 studies focus on point accuracy. For decisions, calibrated uncertainty matters
 134 as well. Conformal prediction [42, 43, 44] provides distribution-free intervals
 135 and has been extended to quantile regression [45], to multi-horizon [46] and
 136 dynamic time series [47], and to arbitrary distribution shift through adaptive
 137 conformal inference [48]. We use these tools to build a calibrated forecaster
 138 and to test, in a controlled way, whether risk-adjusted carbon costs improve

139 scheduling. The question is motivated by the optimizer’s curse [49] and by
 140 the robust-optimisation view that point-estimate solutions can be fragile [50,
 141 51].

142 3. Material and methods

143 3.1. Problem setting

144 We consider an operator that runs a set $\mathcal{S}$ of data-centre sites connected
 145 by a wide-area network. Time is divided into hourly slots $t = 0, 1, 2, \dots$
 146 Deferrable batch jobs arrive online. Job j is released at the start of hour a_j
 147 at origin site $o_j \in \mathcal{S}$. It carries w_j server-hours of work made of independent
 148 tasks (a bag of tasks), can use at most m_j servers concurrently, and must
 149 finish before the start of hour $d_j > a_j$. Work may be paused and resumed,
 150 and any part of it may run at a site other than its origin. Remote execution
 151 requires moving input data. At site s and hour t , $K_s(t)$ servers are free for
 152 batch work once the interactive load has been served.

153 Let $I_s(t)$ be the average carbon intensity of the electricity supplied to site
 154 s during hour t (gCO₂/kWh). Executing one server-hour of job j at site s
 155 during hour t emits

$$c_{o_js}(t) = e \text{PUE}_s I_s(t) + \mathbf{1}[s \neq o_j] \delta \eta \frac{1}{2} (I_{o_j}(t) + I_s(t)), \quad (1)$$

156 where e is the IT energy per server-hour, PUE_s is the power usage effec-
 157 tiveness of site s , δ is the data volume moved per migrated server-hour and
 158 η is the energy intensity of network transfer (kWh/GB). The transfer term
 159 charges network energy at the mean intensity of the two endpoints. With
 160 decision variables $x_{js}(t) \geq 0$ (server-hours of job j run at site s in hour t),
 161 the offline version of the problem is the linear programme

$$\min_{x, u \geq 0} \sum_j \sum_s \sum_{t=a_j}^{d_j-1} c_{o_js}(t) x_{js}(t) + \sum_j \Pi u_j \quad (2)$$

$$\text{s.t.} \quad \sum_s \sum_{t=a_j}^{d_j-1} x_{js}(t) + u_j = w_j \quad \forall j, \quad (3)$$

$$\sum_s x_{js}(t) \leq m_j \quad \forall j, t \in [a_j, d_j), \quad (4)$$

$$\sum_{j: a_j \leq t < d_j} x_{js}(t) \leq K_s(t) \quad \forall s, t, \quad (5)$$

162 where u_j is work left unfinished at the deadline and Π is a lateness penalty
 163 much larger than any emission cost. When migration is not allowed, $x_{js}(t) =$

164 0 for $s \neq o_j$. With complete knowledge of arrivals and intensities, the opti-
 165 mum of (2)–(5) is a lower bound for every online policy, and we use it as the
 166 *clairvoyant oracle*. An online scheduler has to fix $x_s(t)$ at the start of hour
 167 t . At that point it knows the jobs released so far, the current intensities $I_s(t)$
 168 and nothing more than forecasts of $I_s(t+h)$ and of future arrivals.

169 **Proposition 1** (Network-flow structure). *Problem (2)–(5) is a minimum-*
 170 *cost flow problem on a network with a source, one node per job, one node per*
 171 *(job, hour) pair, one node per (hour, site) pair and a sink. Arc capacities*
 172 *are w_j (source $\rightarrow j$), m_j ($j \rightarrow (j, t)$) and $K_s(t)$ ($(t, s) \rightarrow$ sink). The only*
 173 *arcs with a cost are $(j, t) \rightarrow (t, s)$, priced at $c_{o_j s}(t)$, and a slack arc $j \rightarrow$ sink*
 174 *priced at Π .*

175 *Proof.* Conservation at the job node is Eq. (3). The capacity of arc $j \rightarrow$
 176 (j, t) is Eq. (4), conservation at (j, t) splits that hour’s work across sites,
 177 and the capacity of $(t, s) \rightarrow$ sink is Eq. (5). Conversely, every feasible flow
 178 defines a feasible (x, u) with the same cost. The constraint matrix of the
 179 flow formulation is a node-arc incidence matrix, in which every column has
 180 exactly one +1 and one −1, so it is totally unimodular. $\square$

181 Proposition 1 has a practical consequence. The oracle and every plan-
 182 ning problem solved online (Section 3.3) can be solved exactly by network
 183 simplex [32], and integral data (work quantised to 0.01 server-hours) give
 184 integral optimal flows. We use the OR-Tools implementation [52]. On the
 185 instances studied, network simplex returned the same objective value as a
 186 general-purpose LP solver (HiGHS) to within rounding, and it cut the total
 187 run time of the online planners by a factor of about six.

188 3.2. Probabilistic carbon-intensity forecasting

189 *Point model.* For each horizon $h = 1, \dots, H$ ($H = 24$ h) we fit a direct
 190 gradient-boosted regression tree model [53] that predicts the change $I_s(t+h)$
 191 $- I_s(t)$ from information available at issue time t . Direct per-horizon
 192 models avoid the error accumulation of recursive strategies [54]. One model
 193 per horizon is pooled across sites, with the site as a categorical feature. The
 194 features are the current intensity; differences between the current value and
 195 lags of 1, 2, 3, 6, 12 and 23 hours; the value observed at the same hour of
 196 the target on the previous day and in the previous week; a 24-hour rolling
 197 mean and mean absolute change (volatility); the national (GB) intensity;
 198 the regional wind, solar and gas shares; and calendar encodings of the target

time (hour of day, day of week, annual sine and cosine). Models are fitted with absolute-error loss, so they estimate the conditional median. We use scikit-learn’s histogram gradient boosting [55] with 400 iterations, learning rate 0.05, 31 leaves and at least 50 samples per leaf. The models use only information available when the forecast is issued, and no weather forecasts. This is deliberate: the goal is a forecaster any operator can build from a public intensity feed.

Conformal upper quantiles. A point forecast $\hat{I}_s(t+h|t)$ is turned into calibrated upper quantiles by split conformal prediction [42, 43] with normalised scores. On a calibration window disjoint from training, we compute for each site s and horizon h

$$r_{s,h}(t) = \frac{I_s(t+h) - \hat{I}_s(t+h|t)}{\sigma_s(t) + \epsilon}, \quad (6)$$

where $\sigma_s(t)$ is the 24-hour mean absolute hourly change observed at issue time and $\epsilon = 5$ gCO₂/kWh. For a target level β , the upper forecast is $U_s^\beta(t+h|t) = \hat{I}_s(t+h|t) + \hat{q}_{s,h}(\beta_{s,h}(t))(\sigma_s(t) + \epsilon)$, where $\hat{q}_{s,h}(\cdot)$ is the empirical quantile function of the calibration scores. Grid carbon intensity is not exchangeable over time: 2024 was markedly cleaner than 2022–2023 (Fig. 3). We therefore update the working level online by adaptive conformal inference (ACI) [48]. Once $I_s(t+h)$ is observed,

$$\alpha_{s,h} \leftarrow \alpha_{s,h} + \gamma((1-\beta) - \mathbf{1}[I_s(t+h) > U_s^\beta(t+h|t)]), \quad \beta_{s,h}(t) = 1 - \alpha_{s,h}, \quad (7)$$

with step size $\gamma = 0.005$. By the ACI result of [48], the long-run frequency with which realised intensities exceed U^β converges to $1 - \beta$ for any data sequence, with no distributional assumption. This matters for the scheduler: the level β becomes an interpretable risk dial, and it stays valid under drift. With $\beta = 0.5$ the procedure also removes level shifts in the point model (a recalibrated median).

3.3. CARMA: carbon-aware anticipatory receding-horizon multi-site allocation

A standard receding-horizon (model predictive control, MPC) scheduler [35] solves, at every hour t , the flow problem of Proposition 1 restricted to pending jobs and the window $[t, t+H)$. Future intensities are replaced by forecasts, and only the first hour of the plan is executed. Such a planner is *myopic*

229 *about demand*: it sees only jobs that have already arrived. The next result
 230 shows that this can be arbitrarily costly even when carbon intensities are
 231 known exactly.

232 **Proposition 2** (Myopia can be unboundedly costly). *Even with perfect in-*
 233 *tensity forecasts, myopic receding-horizon scheduling has no finite competitive*
 234 *ratio with respect to the clairvoyant oracle.*

235 *Proof.* Take a clean site A with one server and a dirty site B with ample
 236 capacity. Let $c_A(0) = \varepsilon$, $c_A(1) = 0$ and $c_B(0) = c_B(1) = c$, and ignore
 237 transfer costs. A flexible job ($w = 1$, $m = 1$) arrives at $t = 0$ with deadline
 238 $d = 2$. An urgent job ($w = 1$) arrives at $t = 1$ with $d = 2$. At $t = 0$ the
 239 myopic planner sees only the flexible job and defers it to A at $t = 1$ (cost
 240 $0 < \varepsilon$). At $t = 1$ both jobs must run in the same hour, one of them at B , so
 241 the total cost is c . The oracle runs the flexible job at $t = 0$ and the urgent
 242 job at $t = 1$, both at A , for a total of ε . The ratio c/ε is unbounded as
 243 $\varepsilon \rightarrow 0$. $\square$

244 The mechanism in Proposition 2 is common in practice. Low-carbon
 245 capacity is scarce, and a myopic planner spends it on flexible work that
 246 could have waited or run elsewhere, leaving urgent jobs that arrive later
 247 to run on carbon-intensive capacity. CARMA addresses it by anticipating
 248 demand inside the same flow problem, and optionally prices forecast risk.

249 *Anticipated demand (ghost jobs).* From the operator's own history we esti-
 250 mate, for every hour of the week τ , origin o and window class b , the expected
 251 work $\bar{W}_{\tau ob}$ and the expected total width $\bar{M}_{\tau ob}$ of arriving jobs. Window classes
 252 are the bins $\mathcal{B} = \{1, 2, 4, 6, 9, 13, 18, 24\}$ hours. A job is assigned to the largest
 253 bin not exceeding its window, so ghost jobs are never more flexible than
 254 the jobs they represent. At hour t , for each $k = 1, \dots, \ell$ (the look-ahead),
 255 CARMA adds to the planning problem a *ghost job* for every (o, b) with work
 256 $\kappa \bar{W}_{\tau(t+k)ob}$, width $\kappa \bar{M}_{\tau(t+k)ob}$, release $t + k$ and deadline $\min(t + k + b, t + H)$.
 257 Ghost jobs compete with real jobs for capacity, but their slack arc is priced
 258 at a moderate penalty Π_g , set to 1.5 times the largest local unit cost in the
 259 window, rather than at Π . They can therefore reserve clean capacity when it
 260 is economical but never force real work to be late. The weight $\kappa \geq 0$ scales
 261 how much of the expected demand is anticipated. With $\kappa = 0$, CARMA
 262 reduces to myopic MPC.

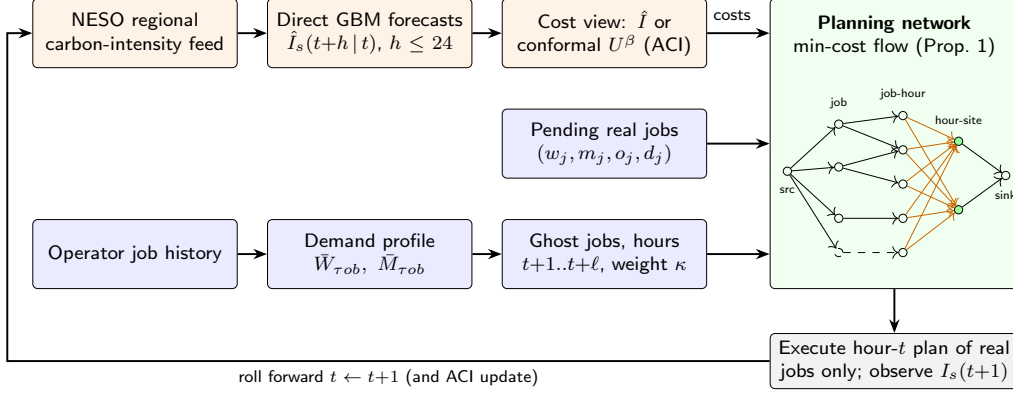


Figure 1: The CARMA decision loop. Every hour, carbon-intensity forecasts price the (job-hour) $\rightarrow$ (hour-site) arcs, and pending real jobs and ghost jobs for expected arrivals (dashed) share the planning network. The min-cost flow is solved exactly, and only the first-hour allocation of real jobs is executed.

263 *Carbon cost view.* Future-hour costs in Eq. (1) are evaluated at a forecast of
 264 $I_s(t+h)$; for the current hour ($h = 0$) the observed intensity is used. CARMA
 265 admits two cost views: the point forecast $\hat{I}$, or the conformal upper quantile
 266 U^β of Section 3.2. The second is motivated by the optimizer’s curse [49]:
 267 selecting the minimum over noisy estimates is biased towards optimism, and a
 268 level $\beta > 0.5$ penalises uncertain future hours in proportion to their calibrated
 269 spread. Whether this risk premium helps is an empirical question, and we
 270 treat the cost view as a hyperparameter.

271 *Algorithm.* Fig. 1 summarises the loop. At each hour t , CARMA (i) observes
 272 $I_s(t)$, the free capacity and the pending jobs; (ii) builds the cost cube $c_{os}(t +$
 273 $h)$ for $h < H$ from the chosen forecast view; (iii) adds the ghost jobs for
 274 hours $t + 1, \dots, t + \ell$; (iv) solves the resulting min-cost flow problem exactly;
 275 (v) executes the first-hour allocations of the real jobs only; and (vi) after
 276 observing the realised intensities, updates the ACI levels by Eq. (7) when
 277 conformal costs are used. Overdue work gets a one-hour window and is
 278 served with priority. The planning network has $O((n_t + \ell|\mathcal{S}||\mathcal{B}|)H|\mathcal{S}|)$ arcs,
 279 where n_t is the number of pending jobs, so each decision takes a few tens of
 280 milliseconds (Section 5). The hyperparameters (κ, ℓ) and the cost view are
 281 tuned on validation weeks that precede the test period (Section 5.2).

282 3.4. Baselines

283 We compare CARMA with the following online policies and the oracle.
 284 **ASAP-Local** is carbon-agnostic: it runs work at its origin as soon as pos-
 285 sible, earliest deadline first (EDF). **Greedy-Spatial** also runs work imme-
 286 diately, but at the site that is currently cleanest; this is geographical load
 287 balancing without temporal shifting [13]. **Forecast-Reserve** books, when
 288 a job arrives, the cheapest forecast (site, hour) slots in its window that still
 289 have capacity, and never revises the booking. This is the lowest-window
 290 strategy of [10] extended to multiple sites. **MPC** is the myopic receding-
 291 horizon flow scheduler with point forecasts. **MPC+conformal** uses the
 292 conformal upper quantile at the level that was best for MPC on validation
 293 data. **MPC-Perfect** is MPC given the realised intensities, which isolates
 294 the value of forecast accuracy. **CARMA-Perfect** is CARMA given the
 295 realised intensities, and **CARMA+conformal** is CARMA with the best
 296 conformal cost view. **Oracle** solves (2)–(5) with full knowledge of arrivals
 297 and intensities.

298 3.5. Data and experimental design

299 *Carbon intensity.* We use half-hourly regional carbon intensity for Great
 300 Britain from the Carbon Intensity API of the National Energy System Op-
 301 erator (NESO) [41] from 1 January 2022 to 31 December 2024. Values are
 302 averaged to hourly resolution. Four regions host the candidate sites: South
 303 Scotland, North West England, West Midlands and London. They span the
 304 north–south gradient of the British grid, from wind- and nuclear-rich South
 305 Scotland to gas-dependent London, and include the main data-centre mar-
 306 kets of London, Manchester (North West England) and Birmingham (West
 307 Midlands). Gaps (602 of 26,304 hours in at least one region) were filled by
 308 linear interpolation. The period up to 30 June 2023 is used to train the
 309 forecaster, July–September 2023 to calibrate the conformal scores, October–
 310 December 2023 to tune the scheduler, and all 52 weeks starting on Mondays
 311 in 2024 as the out-of-sample test set.

312 *Sites and energy parameters.* Batch capacities are 300, 400, 400 and 600
 313 servers for South Scotland, North West England, West Midlands and Lon-
 314 don. This reflects a small, clean northern site and a large site in London.
 315 Interactive load takes $20\% \pm 15\%$ of each site’s capacity on a diurnal cy-
 316 cle that peaks in the afternoon. We assume $e = 0.4$ kWh per server-hour.
 317 The assumed PUE values are 1.15, 1.20, 1.25 and 1.30 from north to south,

reflecting free-cooling potential. Migrated work moves $\delta = 2$ GB per server-hour at $\eta = 0.02$ kWh/GB. This value lies within the wide range of published estimates of transmission energy intensity [56]. We vary η in a sensitivity analysis.

Workloads. Job arrivals follow a non-homogeneous Poisson process with a diurnal peak in the afternoon and 30% lower weekend rates. Jobs originate at the four sites with probabilities 0.10, 0.20, 0.25 and 0.45. Width m_j is log-uniform on $[4, 64]$ servers and the run time at full width r_j is uniform on $\{1, \dots, 6\}$ hours, so $w_j = m_j r_j$. A share v of jobs is urgent (deadline $= a_j + r_j$, no temporal slack); for the others the window is $\lceil f_j r_j \rceil$ hours with $f_j \sim U[1.5, 4]$, capped at 24 hours. The baseline value is $v = 0.3$. The arrival rate is scaled so that the offered load equals a fraction $\rho \in \{0.3, 0.5, 0.7\}$ of the mean free capacity. Each test episode covers one week of arrivals. We also simulate 24 hours of arrivals before and after the week, so the evaluated week starts and ends under steady-state congestion. Metrics are computed only for jobs released during the week. For each week and load level we draw one workload trace with a fixed seed, and every policy runs on that same trace. The demand profile used by CARMA is estimated from 13 independent historical traces. We study two regimes: *spatio-temporal*, where migration is allowed, and *temporal-only*, where it is not.

Metrics and statistics. The primary metric is operational emissions (tCO₂) per week. We report the reduction relative to ASAP-Local and the optimality gap relative to the oracle, the share of work finished late, the share of work executed away from its origin, and the decision time. Policies are compared by paired two-sided Wilcoxon signed-rank tests over the 52 test weeks, with Holm correction across comparisons.

4. Theoretical guarantees

This section establishes three kinds of guarantees. The first is a worst-case limit that applies to *every* online scheduler. The second is a set of worst-case and prediction-dependent guarantees for CARMA. The third is a fairness property of the carbon attribution induced by the pooled scheduling problem. Proofs are given in Appendix A. Throughout, unit costs (Eq. (1), including any backstop resource introduced below) satisfy $0 < L \leq c_{os}(t) \leq U$, and $\theta = U/L$ denotes the *carbon-intensity ratio*. A positive floor L holds as soon as the embodied emissions of servers, or a non-zero minimum grid intensity,

are counted. An online algorithm is ρ -competitive if its emissions never exceed ρ times those of the clairvoyant oracle, for any arrival sequence and any intensity trajectory.

4.1. A universal lower bound under demand uncertainty

Competitive analysis of carbon-aware scheduling has so far concentrated on uncertainty about future *prices*, that is, future intensities, for a single workload. Examples are online search and one-way trading [22], the online pause-and-resume problem [23] and spatiotemporal allocation with deadlines [24]. When several jobs share scarce low-carbon capacity, a second source of difficulty appears: uncertainty about *future demand*. The next result shows that this uncertainty alone rules out good worst-case performance, however accurate the carbon forecasts are.

Theorem 1 (Lower bound). *Let $\theta \geq 1$. No online algorithm (deterministic or randomised, with integral or fractional decisions) can be better than $R^*(\theta)$ -competitive, even if it knows all future carbon intensities exactly, where*

$$R^*(\theta) = \max_{1 \leq p \leq \theta} \left\{ 1 + \frac{(p-1)(\theta-p)}{p^2 - p - 1 + \theta} \right\} \geq 1 + \frac{\sqrt{\theta}(\sqrt{\theta}-1)^2}{2\theta - \sqrt{\theta} - 1} = \frac{\sqrt{\theta}}{2} + O(1). \quad (8)$$

On the same instance, myopic receding-horizon scheduling (MPC) has ratio at least $(1 + \theta)/2$.

The bound is attained on a two-hour instance, a quantitative version of Proposition 2. Any algorithm must decide how much flexible work to run now at moderate cost before it learns whether an urgent job will claim the cheap capacity later. For example, $R^*(100) = 5.29$ and $R^*(1000) = 16.1$, whereas the corresponding myopic ratios are 50.5 and 500.5. Theorem 1 also explains the empirical finding of Section 5.3: carbon forecasts cannot substitute for information about demand. It motivates the learning-augmented design of CARMA [25], in which predictions of demand are used when they are good and cannot cause unbounded harm when they are bad.

4.2. Robustness, consistency and smoothness of CARMA

We analyse CARMA under two assumptions.

(A1) *Backstop capacity.* In every hour, work may also run on a backstop resource of unlimited capacity at a unit cost in $[L, U]$. Examples are

383 bursting to a public-cloud region or to reserve servers at a carbon-
 384 intensive site. All jobs, real and predicted, are *window-feasible*, $w_j \leq$
 385 $m_j(d_j - a_j)$. The penalties satisfy $\Pi > U$ for real jobs and $\Pi_g \geq U$ for
 386 ghost jobs. The implementation satisfies the latter, because Π_g is 1.5
 387 times the largest local unit cost, which exceeds every unit cost including
 388 transfer overheads.

389 (A2) *Episode within the planning window.* The episode $\{0, \dots, T - 1\}$ has
 390 length $T \leq H$, as for a daily batch window, so every planning problem
 391 extends to the end of the episode.

392 To measure prediction quality at hour t , let F_t be the set of jobs that actually
 393 arrive after t , and $\hat{F}_t$ the ghost jobs used by CARMA. Jobs are compared
 394 after splitting them into *atoms* with proportional widths, which leaves every
 395 planning problem unchanged (Appendix A). Let d_t be the smallest total
 396 work of atoms that must be added to or removed from $\hat{F}_t$ to obtain F_t . Let
 397 $\varepsilon_t = \max_{\tau > t, o, s} |\hat{c}_{os}(\tau | t) - c_{os}(\tau)|$ be the carbon-cost forecast error, and W_t
 398 the total work pending at t plus the larger of the true and predicted future
 399 work.

400 **Theorem 2** (Guarantees for CARMA). *Under (A1), for any arrival se-*
 401 *quence and any predictions, CARMA completes every job by its deadline and*
 402 *is θ -competitive (robustness). If (A2) also holds, then*

$$\text{ALG} \leq \min \left\{ \theta \cdot \text{OPT}, \text{OPT} + \sum_{t=0}^{T-1} [(U - L) d_t + 2 \varepsilon_t W_t] \right\}. \quad (9)$$

403 *In particular, with exact predictions ($d_t = \varepsilon_t = 0$) CARMA is optimal*
 404 *(consistency), and its excess emissions grow at most linearly in the prediction*
 405 *errors (smoothness).*

406 The three properties are the standard desiderata for algorithms with pre-
 407 dictions [25, 24]. Two features of the bound matter in practice. First, demand
 408 errors are paid at rate $U - L$, the spread of unit costs, not at the lateness
 409 penalty. This is because the backstop absorbs any misprediction, so a wrong
 410 prediction can make CARMA use dirtier capacity but can never make a real
 411 job late. Second, the bound does not depend on how the forecasts were pro-
 412 duced, so it holds for the point and conformal cost views alike. Without
 413 demand predictions, myopic MPC has no consistency guarantee: by Theo-
 414 rem 1 it can be a factor of $(1 + \theta)/2$ from optimal even with perfect carbon

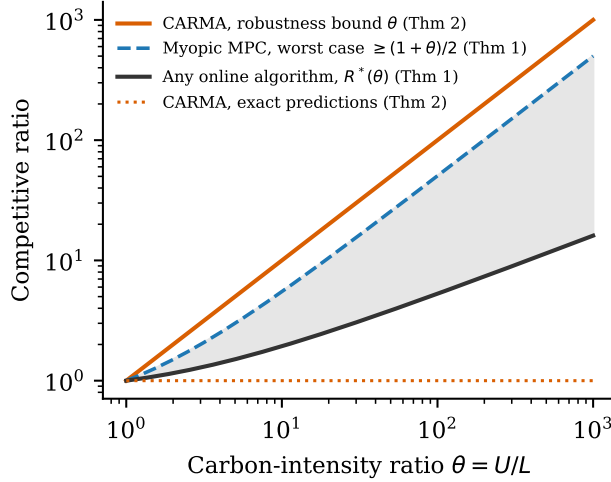


Figure 2: Worst-case landscape as a function of the carbon-intensity ratio θ . No online algorithm can lie below $R^*(\theta)$ (Theorem 1). Myopic MPC can be driven to $(1 + \theta)/2$ even with perfect carbon forecasts. CARMA is never worse than θ (given a backstop) and is optimal when its predictions are exact (Theorem 2).

forecasts. Robustness θ is not optimal in view of the $\Theta(\sqrt{\theta})$ lower bound. Closing this gap, for example by protecting deferral decisions with reservation prices in the spirit of [22, 24], is left for future work. Assumption (A2) is needed only for the smoothness term, because horizon truncation couples decisions beyond the planning window. The experiments in Section 5.5 use weekly episodes with $H = 24$ hours and test the degradation empirically. Fig. 2 summarises the worst-case landscape.

4.3. Fair carbon attribution among sites

Pooling capacity reduces total emissions, but it redistributes them. Work from London is executed in South Scotland, and Scottish jobs may be displaced by it. The organisations that operate the sites (or the tenants that submit the jobs) therefore need an attribution of the pooled emissions that each of them can accept. We model this as a cooperative game. The players are the sites $\mathcal{S}$. A coalition $C \subseteq \mathcal{S}$ pools the capacity of its sites and serves the jobs submitted at them. Its stand-alone value $c(C)$ is the optimum of (2)–(5) restricted to those jobs and sites. An attribution $\phi \in \mathbb{R}^{|\mathcal{S}|}$ is *budget-balanced* if $\sum_o \phi_o = c(\mathcal{S})$. It is in the *core* if $\sum_{o \in C} \phi_o \leq c(C)$ for every

432 coalition C . A core attribution gives no group of sites a reason to leave the
 433 pool and schedule on its own.

434 Let (λ, ν, μ) be optimal dual variables of the grand-coalition problem: λ_j
 435 for the work constraint (3), $\nu_{jt} \leq 0$ for the width constraint (4) and $\mu_{st} \leq 0$
 436 for the capacity constraint (5). Define the *dual carbon attribution*

$$\phi_o = \sum_{j: o_j=o} \left(\lambda_j w_j + \sum_t \nu_{jt} m_j \right) + \sum_t \mu_{ot} K_o(t). \quad (10)$$

437 Site o is charged the carbon shadow price of the work it submits and is
 438 credited ($\mu \leq 0$) with the shadow value of the capacity it contributes.

439 **Theorem 3** (Core-stable carbon attribution). *The attribution (10) is budget-*
 440 *balanced and in the core of the pooling game. In particular, it is individually*
 441 *rational: no site is charged more than its stand-alone optimum $c(\{o\})$. It is*
 442 *obtained from the node potentials of the min-cost flow of Proposition 1, at no*
 443 *extra computational cost.*

444 The result follows from Owen’s theorem on linear production games [27].
 445 Games defined by network flows are known to be totally balanced [28]. Car-
 446 bon reporting is done after the fact, so the hindsight game is the relevant
 447 object for accounting. The simpler *location-based* attribution, in which each
 448 site is charged the emissions of its own jobs wherever they ran, carries no
 449 such guarantee. Section 5.6 shows that in practice it violates the core in
 450 every test week, whereas the dual attribution never does for the hindsight
 451 benchmark.

452 5. Results

453 5.1. Carbon-intensity data and forecasts

454 Fig. 3 shows the regional intensities. Over 2022–2024, mean intensity was
 455 37 gCO₂/kWh in South Scotland, 58 in North West England, 156 in the West
 456 Midlands and 163 in London. At any hour the gap between the cleanest and
 457 dirtiest site is therefore large, which is what makes spatial shifting attractive.
 458 Intensities fell steadily: London’s annual mean dropped from 202 gCO₂/kWh
 459 in 2022 to 124 in 2024. Diurnal profiles peak in the early evening (around
 460 18:00 UTC), and the profile of South Scotland is much flatter than those
 461 of the southern sites. The four series are positively correlated (pairwise

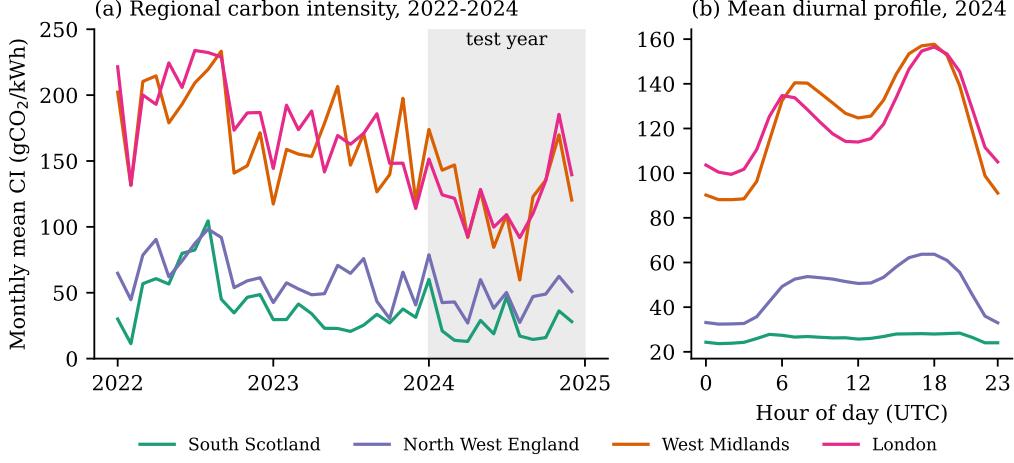


Figure 3: Hourly carbon intensity of the four candidate sites (NESO regional data). (a) Monthly means, 2022–2024; the shaded area is the out-of-sample test year. (b) Mean diurnal profile in 2024 (UTC).

462 correlations between 0.50 and 0.87), because national wind output drives all
 463 of them.

464 Table 1 and Fig. 4a summarise forecast accuracy in 2024. The forecaster
 465 was trained on 1.24 million (issue time, site, horizon) samples in about one
 466 minute. Averaged over horizons and sites, the direct GBM forecaster has a
 467 mean absolute error (MAE) of 31.2 gCO₂/kWh, against 34.4 for persistence
 468 and 40.4 for the seasonal-naïve forecast. The advantage is largest at short
 469 and medium horizons: MAE falls by 23% at 1 hour (6.2 against 8.0) and
 470 by 23% at 6 hours (23.3 against 30.2). Beyond about 18 hours, all methods
 471 converge to the seasonal-naïve level, and at 24 hours the GBM is slightly
 472 worse than the seasonal-naïve benchmark. Without weather forecasts, day-
 473 ahead changes in wind output are hard to predict from the intensity history
 474 alone [40]. By site, the GBM is most accurate for the volatile southern
 475 sites. In South Scotland, where intensity is low and often flat, persistence is
 476 marginally better.

477 Calibration matters more than point accuracy for the risk-adjusted cost
 478 view, and Fig. 4b shows the effect of distribution shift. Static split-conformal
 479 bounds, calibrated on July–September 2023, under-cover in 2024 at every
 480 level: at $\beta = 0.9$, annual coverage was 0.858 and monthly coverage ranged
 481 from 0.68 to 0.95. With ACI, annual coverage matches the target to within

Table 1: Forecast accuracy on the 2024 test year (MAE in gCO₂/kWh; mean over sites unless stated).

Model	Horizon h (hours)					All horizons	
	1	3	6	12	24	MAE	RMSE
Persistence	8.0	19.3	30.2	37.4	40.4	34.4	48.4
Seasonal naive (24 h)	40.4	40.4	40.4	40.4	40.4	40.4	57.0
Direct GBM (this study)	6.2	14.9	23.3	33.9	42.4	31.2	41.0

MAE by site over all horizons (persistence / GBM): South Scotland 14.1 / 16.2; North West England 24.7 / 21.8; West Midlands 56.1 / 49.3; London 42.7 / 37.6.

0.002 at all levels (0.500, 0.599, 0.698, 0.798 and 0.898). Monthly coverage stays within ± 4 percentage points of the target, and coverage is uniform across horizons. At $\beta = 0.5$, the ACI-adjusted median lies on average 9.9 gCO₂/kWh below the point forecast, which reflects the downward drift in intensity that the static model does not capture.

5.2. Hyperparameter tuning

Fig. 5 reports the validation results (13 weeks, October–December 2023, $\rho \in \{0.3, 0.5, 0.7\}$, migration allowed). Anticipating demand helps across the whole grid: every tested configuration beats myopic MPC, whose mean gap to the oracle is 20.3%. The gap is smallest around $\kappa = 1$, where expected demand is anticipated without inflation. For $\ell = 12$ it falls to 7.3%, and for $\ell = 18$ to 7.2%. Over-anticipation ($\kappa = 1.25$ with $\ell = 18$) reserves too much clean capacity and is penalised. We selected $\kappa = 1$ and $\ell = 12$ hours. Configurations within 0.25 percentage points of the best were resolved towards the shorter look-ahead, which keeps the network smaller.

The cost view shows a clear interaction with anticipation (Fig. 5b). For myopic MPC, pessimistic conformal costs help substantially. The gap falls monotonically from 23.1% at $\beta = 0.5$ to 16.2% at $\beta = 0.8$, well below the 20.3% obtained with point forecasts. For CARMA, every conformal level is worse than the point forecast: the best level, $\beta = 0.6$, gives 8.2% against 7.3%. The explanation is that a risk premium on future hours makes the planner run work now rather than later, which keeps future clean capacity free. For a myopic planner this is a crude form of anticipation. Once CARMA represents future demand explicitly, the same premium double-counts it and gives up useful deferrals. The selected CARMA therefore uses point forecasts.

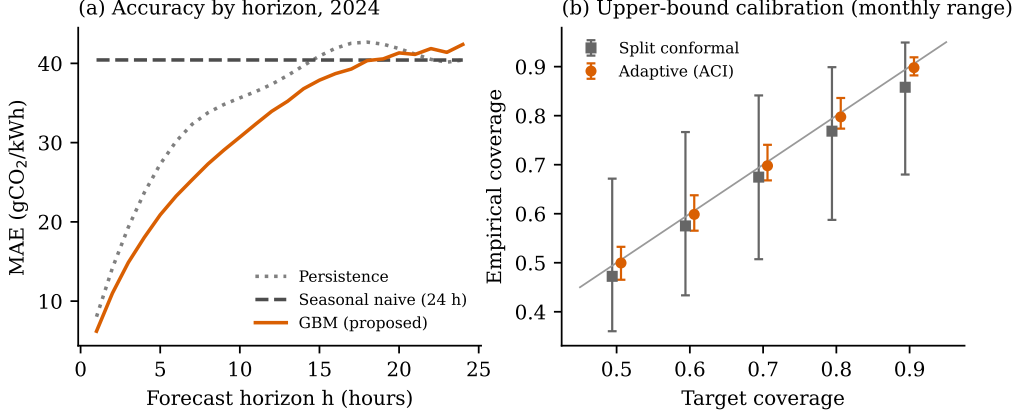


Figure 4: Forecast evaluation in 2024. (a) MAE by horizon, averaged over sites. (b) Empirical coverage of one-sided upper bounds U^β against the target level β for static split conformal calibration and adaptive conformal inference (ACI). Markers show the annual coverage and bars the range across the 12 months.

507 For the test we keep MPC+conformal at $\beta = 0.8$ and CARMA+conformal
 508 at $\beta = 0.6$ as ablations.

509 5.3. Out-of-sample scheduling performance

510 Tables 2 and 3 report results for the 52 test weeks of 2024. Weekly traces
 511 contain on average 920, 1,529 and 2,144 jobs (69,900, 115,600 and 162,600
 512 server-hours) at $\rho = 0.3, 0.5$ and 0.7 . Fig. 6 shows the gaps to the oracle
 513 with 95% confidence intervals.

514 *Spatio-temporal regime.* Carbon-aware scheduling gives large reductions when
 515 migration is allowed. The oracle cuts emissions relative to ASAP-Local by
 516 70.5%, 56.7% and 39.5% at $\rho = 0.3, 0.5$ and 0.7 . The achievable reduc-
 517 tion shrinks with load because less clean capacity is left per unit of work.
 518 Greedy-Spatial realises only part of this potential (gaps of 35–65%), because
 519 it has no temporal flexibility and floods the cleanest site with whatever ar-
 520 rives first. Forecast-Reserve and MPC come closer, with gaps of 18–26%.
 521 CARMA reduces emissions by 68.1%, 53.3% and 36.1% and reaches gaps of
 522 8.7%, 8.2% and 5.8%. Relative to MPC this is a further emission reduction
 523 of 7.6%, 9.4% and 8.8%, and the gap to the optimum shrinks by a factor
 524 of 2.2–3.0. CARMA’s emissions are lower than MPC’s in 51, 52 and 52 of
 525 the 52 weeks, and lower than Forecast-Reserve’s in every week (Fig. 7b). All

Table 2: Out-of-sample results in the spatio-temporal regime (migration allowed), 52 weeks of 2024. Emis.: mean weekly operational emissions (tCO₂). Red.: reduction relative to ASAP-Local (%). Gap: excess over the clairvoyant oracle (%). Late: share of work finished after its deadline (%). Mig.: share of work executed away from its origin (%). p : Holm-adjusted two-sided Wilcoxon signed-rank test of weekly emissions against CARMA. The best gap among implementable online policies is in bold.

Load	Policy	Emis.	Red.	Gap	Late	Mig.	p
$\rho = 0.3$	ASAP-Local	3.73	0.0	274.7	0.00	0.0	<0.001
	Greedy-Spatial	1.77	53.8	64.6	0.00	81.6	<0.001
	Forecast-Reserve	1.40	63.7	25.7	0.00	83.7	<0.001
	MPC	1.34	65.5	18.9	0.00	69.2	<0.001
	MPC+conformal	1.29	66.7	14.2	0.00	70.0	<0.001
	MPC-Perfect	1.30	66.4	15.9	0.00	68.9	<0.001
	CARMA	1.23	68.1	8.7	0.00	69.9	–
	CARMA+conformal	1.24	68.1	8.9	0.00	70.2	0.122
	CARMA-Perfect	1.17	70.0	1.8	0.00	69.6	<0.001
	Oracle	1.15	70.5	0.0	0.00	70.0	–
$\rho = 0.5$	ASAP-Local	6.12	0.0	142.7	0.05	0.0	<0.001
	Greedy-Spatial	4.15	33.3	58.8	0.00	76.3	<0.001
	Forecast-Reserve	3.36	46.6	25.0	0.02	78.2	<0.001
	MPC	3.27	48.1	21.4	0.01	50.0	<0.001
	MPC+conformal	3.18	49.7	17.4	0.00	50.7	<0.001
	MPC-Perfect	3.22	48.9	19.5	0.02	49.4	<0.001
	CARMA	2.96	53.3	8.2	0.00	50.3	–
	CARMA+conformal	2.98	53.2	8.6	0.00	50.4	0.001
	CARMA-Perfect	2.83	55.7	2.3	0.00	50.0	<0.001
	Oracle	2.77	56.7	0.0	0.00	50.7	–
$\rho = 0.7$	ASAP-Local	8.32	0.0	68.1	6.50	0.0	<0.001
	Greedy-Spatial	6.79	19.1	35.3	0.00	73.4	<0.001
	Forecast-Reserve	6.15	26.9	21.7	1.16	73.6	<0.001
	MPC	5.95	29.3	17.6	2.00	31.6	<0.001
	MPC+conformal	5.85	30.7	15.3	0.78	32.1	<0.001
	MPC-Perfect	5.89	30.1	16.3	2.08	31.4	<0.001
	CARMA	5.43	36.1	5.8	0.05	31.5	–
	CARMA+conformal	5.45	35.9	6.2	0.02	31.6	<0.001
	CARMA-Perfect	5.26	38.4	1.9	0.05	31.1	<0.001
	Oracle	5.18	39.5	0.0	0.00	30.7	–

Table 3: Out-of-sample results in the temporal-only regime (no migration), 52 weeks of 2024. Columns as in Table 2. Greedy-Spatial is omitted because without migration it coincides with ASAP-Local.

Load	Policy	Emis.	Red.	Gap	Late	p
$\rho = 0.3$	ASAP-Local	3.73	0.0	22.0	0.00	<0.001
	Forecast-Reserve	3.30	11.9	7.1	0.36	<0.001
	MPC	3.26	13.1	5.7	0.01	0.367
	MPC+conformal	3.40	9.2	10.4	0.00	<0.001
	MPC-Perfect	3.11	17.5	0.2	0.02	<0.001
	CARMA	3.26	13.1	5.7	0.00	–
	CARMA+conformal	3.27	12.9	5.9	0.00	0.020
	CARMA-Perfect	3.11	17.6	0.1	0.01	<0.001
	Oracle	3.11	17.7	0.0	0.00	–
$\rho = 0.5$	ASAP-Local	6.12	0.0	17.2	0.05	<0.001
	Forecast-Reserve	5.59	8.9	6.6	3.15	<0.001
	MPC	5.52	10.2	5.1	0.64	0.219
	MPC+conformal	5.67	7.7	8.0	0.10	<0.001
	MPC-Perfect	5.35	13.4	1.3	0.88	<0.001
	CARMA	5.52	10.3	5.0	0.19	–
	CARMA+conformal	5.54	10.1	5.2	0.11	0.003
	CARMA-Perfect	5.31	14.1	0.4	0.26	<0.001
	Oracle	5.29	14.5	0.0	0.01	–
$\rho = 0.7$	ASAP-Local	8.32	0.0	9.5	6.50	<0.001
	Forecast-Reserve	7.96	4.2	4.8	8.35	<0.001
	MPC	7.91	4.9	4.1	12.99	0.244
	MPC+conformal	8.00	3.9	5.2	8.09	<0.001
	MPC-Perfect	7.81	6.3	2.6	15.31	<0.001
	CARMA	7.92	4.8	4.2	6.86	–
	CARMA+conformal	7.95	4.6	4.5	6.72	0.004
	CARMA-Perfect	7.78	6.7	2.1	7.85	<0.001
	Oracle	7.63	8.6	0.0	0.96	–

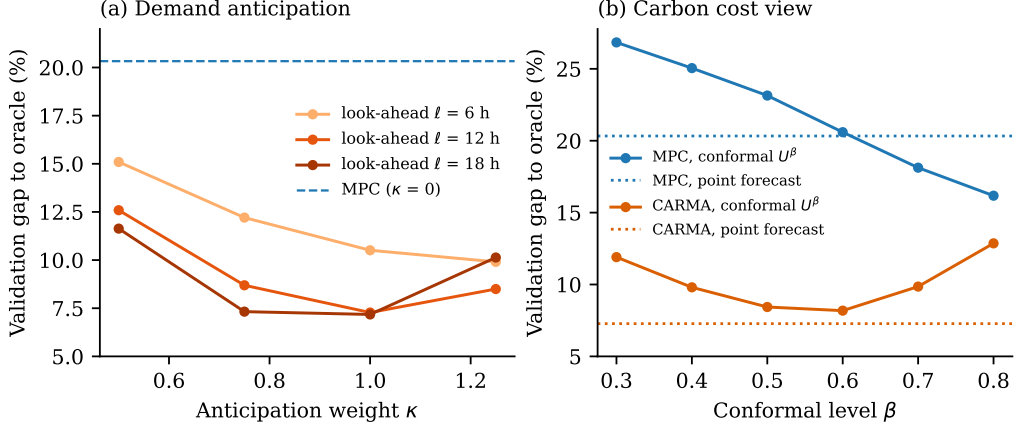


Figure 5: Validation results (2023-Q4, mean over 13 weeks and three load levels). (a) Gap to the oracle as a function of the anticipation weight κ and look-ahead ℓ , using point forecasts. (b) Effect of the carbon cost view: conformal upper quantiles U^β against point forecasts, for myopic MPC and for CARMA ($\kappa = 1$, $\ell = 12$).

526 differences between CARMA and the online baselines are significant after
527 Holm correction ($p < 0.001$). The only exception is CARMA+conformal at
528 $\rho = 0.3$ ($p = 0.12$). CARMA also finishes more work on time under heavy
529 load: at $\rho = 0.7$, 0.05% of its work is late, against 2.0% for MPC, 1.2% for
530 Forecast-Reserve and 6.5% for ASAP-Local. Reserving clean capacity for ur-
531 gent arrivals also reserves capacity in general. CARMA migrates about the
532 same share of work as MPC (31–70%), so its advantage comes from better
533 timing of the use of scarce clean capacity, not from moving more data. For a
534 fleet of 1,700 batch servers at $\rho = 0.5$, the difference between CARMA and
535 MPC is 0.31 tCO₂ per week. The difference from carbon-agnostic operation
536 is 3.16 tCO₂ per week.

537 *Value of information.* The clairvoyant variants separate two sources of loss.
538 With perfect intensity forecasts, the myopic MPC-Perfect still has gaps of
539 15.9%, 19.5% and 16.3%. Perfect carbon foresight is thus worth only 1.3–
540 3.0 percentage points to a myopic planner. Anticipating demand is worth
541 10.3–13.3 points (MPC versus CARMA). With both, CARMA-Perfect comes
542 within 1.8–2.3% of the oracle. The two kinds of information are therefore
543 complementary. Once demand is anticipated, forecast accuracy is worth 3.9–
544 6.8 points (CARMA versus CARMA-Perfect), two to three times more than
545 for the myopic planner. A myopic planner wastes clean capacity whatever

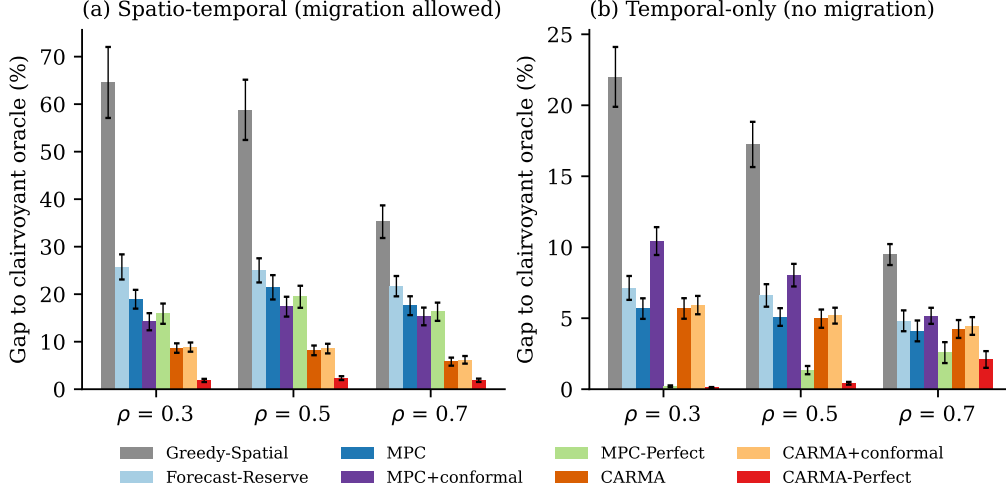


Figure 6: Mean gap to the clairvoyant oracle over the 52 test weeks of 2024 with 95% confidence intervals. ASAP-Local is omitted (gap 68–275% with migration, 9–22% without).

its forecasts, so it cannot benefit from better ones. The weekly results in Fig. 7a show that the ranking is stable across seasons, although the level of achievable reductions varies from week to week with wind conditions.

Temporal-only regime. Without migration (Table 3), each job can only move in time at its origin, and the picture changes. Achievable reductions are much smaller: the oracle saves 17.7%, 14.5% and 8.6%. MPC and CARMA are statistically indistinguishable here ($p \geq 0.22$), both about 4–6% from the oracle, and both are better than Forecast-Reserve. Anticipation brings no benefit because there is no shared pool of clean capacity to protect. Each site’s capacity serves only its own jobs, and the temporal windows of flexible jobs rarely collide with urgent arrivals in a way that deferral cannot absorb. In this regime the remaining loss is due almost entirely to forecast error. MPC-Perfect and CARMA-Perfect close the gap to 0.1–2.6%, and risk-averse conformal costs make MPC worse (gaps of 5.2–10.4%), because they suppress deferrals that would have paid off. Under heavy load ($\rho = 0.7$), temporal shifting also raises lateness: the capacity at the origin sites is itself insufficient, and 6.5% of work is late even under ASAP-Local. CARMA’s late share (6.9%) is close to that level, whereas MPC’s is 13.0%.

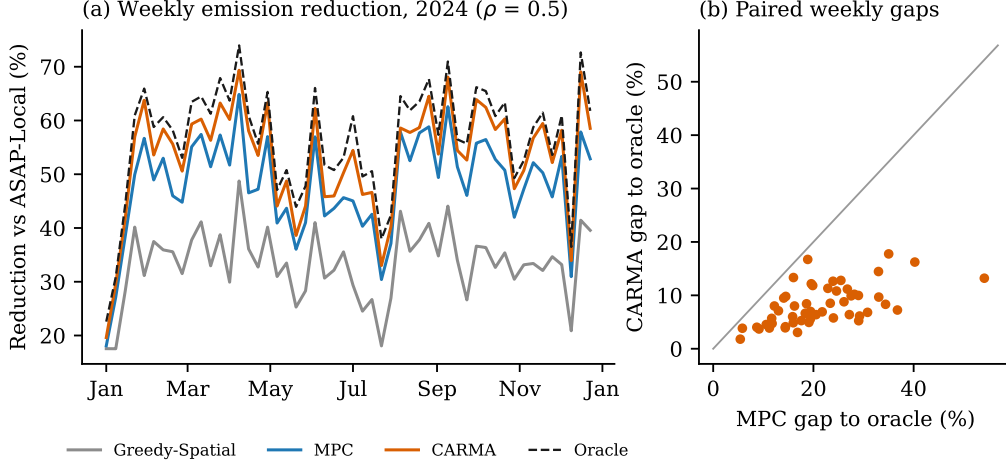


Figure 7: Weekly results in 2024 (spatio-temporal regime, $\rho = 0.5$). (a) Emission reduction relative to ASAP-Local. (b) Paired weekly gaps to the oracle; points below the diagonal are weeks in which CARMA outperforms MPC.

564 *Computation.* A CARMA decision takes 20–47 ms on one CPU core, against
565 2–7 ms for MPC. The difference comes from the ghost jobs, which enlarge
566 the planning network. This is negligible for hourly decisions. A clairvoyant
567 weekly oracle with up to about 2,700 jobs (including warm-up and trailing
568 load) is solved exactly in under one second.

569 5.4. Sensitivity analysis

570 Table 4 varies one factor at a time around the base case: spatio-temporal
571 regime, $\rho = 0.5$, $v = 0.3$, $\eta = 0.02$ kWh/GB. Each setting uses 26 alternating
572 test weeks of 2024 with fresh workload seeds. CARMA keeps its hyperpa-
573 rameters from the base case. CARMA has the smallest gap of all online
574 policies in every setting. Its weekly emissions are below MPC’s in 155 of the
575 156 setting-weeks, and every comparison with a baseline is significant (all
576 $p < 10^{-7}$).

577 *Share of urgent jobs.* The benefit of anticipation grows with the share of
578 urgent work, because urgent arrivals are precisely the demand that myopic
579 planners crowd out. With $v = 0.5$, CARMA’s gap is 5.8% against 20.3% for
580 MPC, and it has no late work (MPC 0.15%). With $v = 0.1$, the gap is 10.4%
581 against 20.9%. When almost all work is flexible, there is less urgent demand
582 to protect capacity for.

Table 4: Sensitivity analysis (spatio-temporal regime; 26 alternating weeks of 2024 per setting; base case $\rho = 0.5$, $v = 0.3$, $\eta = 0.02$ kWh/GB). Oracle red.: emission reduction of the oracle relative to ASAP-Local (%). Gap columns: excess over the oracle (%). Wins: weeks in which CARMA emits less than MPC. Late: share of late work (%), MPC / CARMA.

Setting	Oracle red.	Gap to oracle (%)				Wins vs		Late (%)
		Greedy-Sp.	F.-Reserve	MPC	CARMA	MPC	MPC / CARMA	
Urgent share $v = 0.1$	60.3	74.8	26.3	20.9	10.4	26/26		0.00 / 0.00
Urgent share $v = 0.5$	55.0	51.1	24.5	20.3	5.8	26/26		0.15 / 0.00
Network energy $\eta = 0.005$ kWh/GB	60.7	66.2	26.6	25.0	9.4	26/26		0.01 / 0.00
Network energy $\eta = 0.06$ kWh/GB	52.3	62.4	27.6	18.1	7.4	26/26		0.00 / 0.00
Load $\rho = 0.4$, profile learned at 0.5	65.6	72.8	27.0	21.3	10.8	25/26		0.00 / 0.00
Load $\rho = 0.6$, profile learned at 0.5	49.3	52.2	25.7	22.1	6.9	26/26		0.34 / 0.01

583 *Network energy intensity.* The ranking is unchanged over a twelve-fold range
584 of transmission energy intensity. Cheaper transfers ($\eta = 0.005$ kWh/GB)
585 raise the oracle’s reduction to 60.7%, and CARMA’s gap is 9.4% against
586 25.0% for MPC. Expensive transfers ($\eta = 0.06$ kWh/GB) lower the oracle’s
587 reduction to 52.3%, with gaps of 7.4% and 18.1%. The share of migrated
588 work changes only modestly (49–51% for CARMA), because the intensity
589 difference between South Scotland and London is far larger than any plausible
590 transfer overhead.

591 *Misspecified demand profile.* CARMA’s ghost jobs come from a profile learned
592 at $\rho = 0.5$. When the actual load is 20% higher ($\rho = 0.6$), so that CARMA
593 under-anticipates, its gap is 6.9% against 22.1% for MPC, and its late share
594 is 0.01% against 0.34%. When the actual load is 20% lower ($\rho = 0.4$), so that
595 it over-anticipates, CARMA reserves more clean capacity than needed. Its
596 gap rises to 10.8%, but that is still half of MPC’s 21.3%, and CARMA wins
597 in 25 of the 26 weeks. Anticipation is therefore robust to moderate errors
598 in the demand profile in both directions. The operational safeguard is the
599 softer penalty on ghost jobs: the planner never accepts lateness of real work
600 in exchange for serving ghost demand.

601 5.5. Robustness to prediction errors

602 Theorem 2 predicts that CARMA degrades gracefully as its predictions
603 deteriorate. We test this on every fourth test week of 2024 (13 weeks, $\rho = 0.5$,
604 migration allowed, fresh workload seeds) by injecting controlled errors into
605 the two kinds of prediction (Fig. 8).

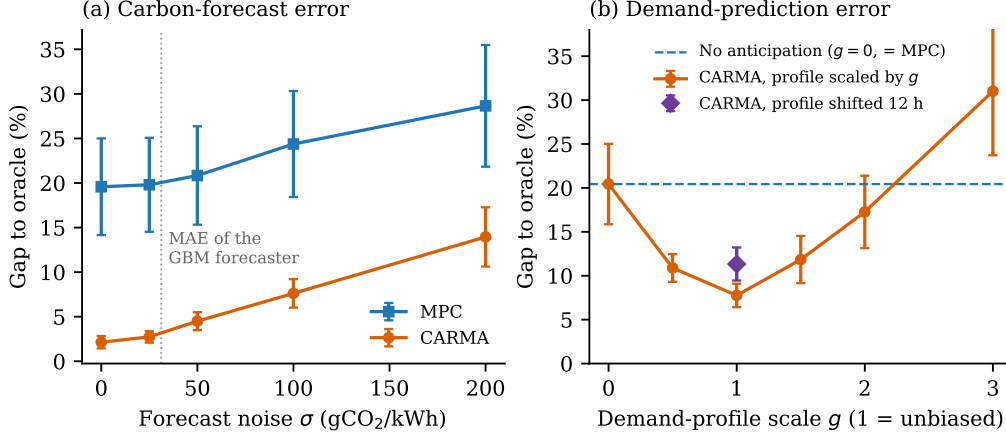


Figure 8: Graceful degradation under controlled prediction errors (13 test weeks of 2024, $\rho = 0.5$, migration allowed; means with 95% confidence intervals). (a) Carbon-intensity forecasts replaced by the realised values plus Gaussian noise of standard deviation σ . (b) Demand profile scaled by g , or shifted by 12 hours so that arrivals are predicted at the wrong time of day.

606 *Carbon-forecast error.* The forecasts were replaced by the realised intensities
607 plus independent Gaussian noise of standard deviation σ . CARMA’s gap to
608 the oracle increases smoothly and almost linearly, from 2.1% with exact fore-
609 casts to 2.7%, 4.5%, 7.6% and 14.0% at $\sigma = 25, 50, 100$ and 200 gCO₂/kWh.
610 The largest level is more than six times the MAE of the GBM forecaster.
611 MPC starts from 19.6% and rises to 28.7%. Even with the noisiest fore-
612 casts, CARMA outperforms MPC with perfect forecasts, which confirms the
613 ordering implied by Theorems 1 and 2.

614 *Demand-prediction error.* Scaling the learned demand profile by g produces
615 a V-shaped response with its minimum at the unbiased profile ($g = 1$, gap
616 7.8%). Under-prediction by half ($g = 0.5$) and over-prediction by half ($g =$
617 1.5) cost 3.1 and 4.1 percentage points. Predicting arrivals at the wrong
618 time of day (a 12-hour shift) costs 3.6 points. Removing anticipation ($g = 0$)
619 returns the MPC level (20.4%). Only gross over-prediction ($g = 3$, triple
620 the true demand) is worse than no anticipation (31.0%), because it makes
621 CARMA hold clean capacity for work that never arrives. Late work was
622 negligible in all 221 runs (at most 0.008% of work). This is consistent with
623 the robustness part of Theorem 2: mispredictions change which capacity is
624 used, but not whether deadlines are met. The largest ratio of CARMA’s

Table 5: Coalitional stability of carbon attributions (52 test weeks of 2024, $\rho = 0.5$). Coalition checks: share of the 52×14 (week, proper coalition) pairs in which the coalition is charged more than its stand-alone emissions. Weeks: share of weeks with at least one violation. IR: share of (week, site) pairs in which a site is charged more than its stand-alone emissions. Max. excess: largest violation (tCO₂).

Benchmark	Attribution	Coalition checks (%)	Weeks (%)	IR (%)	Max. excess
Hindsight optimum	Location-based	46.4	100.0	49.5	1.25
	Work-proportional	44.4	100.0	45.7	1.43
	Dual (Theorem 3)	0.0	0.0	0.0	–
CARMA (realised)	Location-based	43.0	100.0	44.2	1.21
	Work-proportional	43.1	100.0	46.2	1.47
	Dual (Theorem 3)	0.7	9.6	0.0	0.06

emissions to the oracle’s over all runs was 1.52, far below the worst-case bound θ .

5.6. Fair carbon attribution

We evaluate Theorem 3 on the 52 test weeks at $\rho = 0.5$, treating the four sites as players. For every week and each of the 15 coalitions, we compute the hindsight value $c(C)$, and we also run CARMA with only the coalition’s sites and jobs. Pooling pays: the hindsight optimum of the grand coalition emits 3.37 tCO₂ per week, against 6.70 for the four sites operating separately. With CARMA the figures are 3.63 against 6.88. The question is how to share the reduction. We compare three attributions. The *location-based* attribution charges each site the emissions of its own jobs wherever they ran. The *work-proportional* attribution splits the total in proportion to submitted work. The *dual* attribution is Eq. (10). For CARMA’s realised emissions, the dual attribution is applied to the hindsight value, and the online excess over the hindsight optimum is split in proportion to work. A check is violated when a coalition is charged more than it would emit on its own.

Table 5 and Fig. 9 report the results. In the hindsight game, the dual attribution satisfies every one of the 728 coalition checks, as Theorem 3 guarantees. The location-based and work-proportional attributions violate 46% and 44% of the checks and fail in every week. The location-based rule is individually irrational for South Scotland in all 52 weeks and for North West England in 51. Their own jobs are displaced to dirtier capacity by imported work, so under that rule they are charged more than they would emit on their own. The dual attribution resolves this by crediting the clean sites for the capacity they contribute. South Scotland receives a net credit

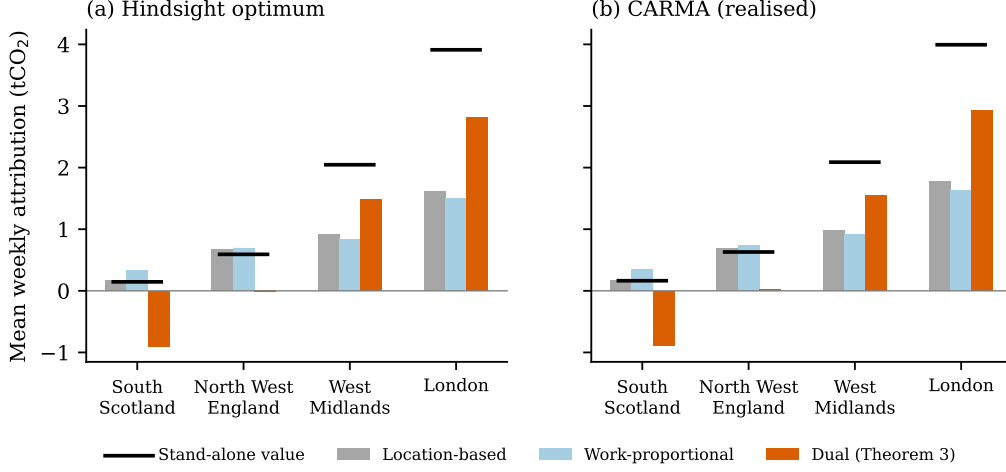


Figure 9: Mean weekly carbon attribution by site under three rules, compared with each site’s stand-alone emissions (black bars), 52 test weeks of 2024 at $\rho = 0.5$. (a) Hindsight optimum. (b) CARMA’s realised emissions. Negative values are net credits for contributing low-carbon capacity.

of 0.91 tCO₂ per week and North West England is close to neutral, while London’s charge (2.81 t) remains well below its stand-alone emissions (3.91 t). With CARMA’s realised emissions, which is the practical case, the adjusted dual attribution remains individually rational for every site in every week. It violates only 5 of the 728 coalition checks, with a largest excess of 0.06 tCO₂, whereas the location-based and work-proportional rules again fail in 43% of the checks. Stand-alone coalitions occasionally leave a little work late (at most 4.7%), and unfinished work emits nothing, so the online checks, if anything, favour secession.

6. Discussion

6.1. Managerial implications

Protect scarce clean capacity. The largest operational loss in multi-site carbon-aware scheduling does not come from imperfect carbon forecasts. It comes from spending low-carbon capacity on work that could have waited or run elsewhere. Proposition 2 and Theorem 1 show this in the worst case, and the 2024 experiments confirm it empirically: with migration allowed, a myopic planner with perfect foresight of carbon intensity is still 16–20% above

667 the optimum. For operators, clean capacity at a low-carbon site should be
 668 treated as a scarce resource with an opportunity cost, much as yield man-
 669 agers protect seats for high-fare late bookings. CARMA sets this protection
 670 level implicitly, through ghost jobs learned from the operator’s own history,
 671 and needs no hand-tuned reservation thresholds.

672 *Invest in forecasts where they pay.* The value of better intensity forecasts
 673 depends on the rest of the policy. For a myopic multi-site planner, perfect
 674 forecasts are worth only 1–3 percentage points. Once demand is anticipated,
 675 they are worth 4–7 points. When work cannot migrate, they are worth al-
 676 most the whole remaining gap. Forecasting effort, for example the integration
 677 of numerical weather predictions [40, 39], should therefore follow, not pre-
 678 ceede, the fix of demand myopia in multi-site settings. In single-site temporal
 679 shifting it is the main lever.

680 *Use risk premia with care.* Calibrated uncertainty is valuable, and ACI kept
 681 coverage on target through a year of strong decarbonisation drift. Its best use
 682 in scheduling is not obvious, however. A pessimistic carbon price improved
 683 a myopic planner by acting as a crude substitute for anticipation, but it
 684 degraded CARMA and hurt temporal-only scheduling, where it suppressed
 685 good deferrals. Because the planner re-optimises every hour, much of the
 686 protection that a risk premium would buy is already provided by recourse.
 687 Calibrated quantiles are better suited to reporting and to service-level deci-
 688 sions than to inflating costs by default.

689 *Guarantees an operator can rely on.* Worst-case properties matter for adop-
 690 tion, because an operator must bound the damage that a wrong forecast
 691 or an unusual week can do. Theorem 2 gives three such bounds. With a
 692 backstop resource, bad predictions can never make a job late. Emissions are
 693 never more than a factor θ above the clairvoyant optimum. Excess emissions
 694 grow at most linearly with prediction error, and the controlled-error exper-
 695 iments confirm this degradation profile. Theorem 1 also sets expectations:
 696 no scheduler can promise near-optimal worst-case emissions without some
 697 knowledge of future demand, so maintaining a demand profile is a necessity,
 698 not a refinement.

699 *Fair carbon accounting for shared capacity.* Pooling halved the emissions of
 700 the four sites, but location-based accounting, which charges each site for its
 701 own jobs wherever they run, would have made South Scotland and North

West England worse off than on their own in almost every week. A multi-organisation or multi-tenant pool attributed this way is unstable, because its cleanest members have an incentive to leave. The dual attribution of Theorem 3 charges each site the carbon shadow price of its demand and credits it for the capacity it contributes. It is budget-balanced, no coalition would prefer to secede, and it is a by-product of the scheduling problem. It offers a principled basis for internal carbon pricing, for sharing the benefits of carbon-aware scheduling among business units, and for tenant-level carbon reporting in shared data centres.

Deployment. CARMA needs three inputs: a public carbon-intensity feed, a demand profile computed from the operator’s own job logs, and a network-flow solver. It decides within tens of milliseconds and returns a plan that is optimal for its model, which makes its decisions auditable. This supports the transparency expected of carbon claims in sustainability reporting. The algorithm is also a rare case in which lower emissions come with better service: under heavy load, CARMA finished far more work on time than the other forecast-driven planners (0.05% late work, against 1.2–2.0%).

6.2. Limitations and future research

The study has several limitations. First, workloads are synthetic. The generator has realistic features (diurnal and weekly seasonality, heavy-tailed job sizes, a mix of urgent and flexible jobs, width limits), but results on production traces may differ. In particular, CARMA’s advantage depends on how predictable arrivals are, and the sensitivity analysis shows how it degrades when the load deviates from the learned profile. Second, we use average (attributional) regional intensities as published by NESO. Marginal emission factors may be more appropriate for evaluating load shifting [6], and regional values in Great Britain are modelled estimates, not metered data. Third, the facility parameters (PUE, network energy) are stylised, idle server power is not modelled, and embodied emissions are outside the scope of this study [20]. Fourth, the forecaster deliberately excludes weather forecasts, so its day-ahead skill is limited. Fifth, bag-of-tasks jobs with free preemption and migration are an idealisation. Jobs with state, data-gravity or data-sovereignty constraints would add restrictions that break the pure network structure. Sixth, the guarantees of Theorem 2 assume a backstop resource and a positive emission floor, and the smoothness bound assumes that the planning window covers the episode. The simulations use neither

738 a backstop nor episodes that short, so they test the predicted behaviour
739 rather than the bound itself. The core property of Theorem 3 is exact for
740 the hindsight benchmark; for CARMA’s realised emissions it held in all but
741 5 of 728 coalition checks.

742 Several extensions follow naturally. The ghost-demand profile could be
743 replaced by probabilistic arrival forecasts, with the anticipation weight set
744 by a newsvendor-type argument. Scenario-based or distributionally robust
745 variants could treat demand and carbon uncertainty jointly [51]. The frame-
746 work can also be coupled with on-site renewables and storage [19, 20] and
747 with demand-response programmes [21]. Finally, the robustness guarantee
748 could be tightened from θ towards the $\Theta(\sqrt{\theta})$ lower bound, for example by
749 protecting deferral decisions with reservation prices [22, 24] while retaining
750 consistency, and the attribution could be extended to online, per-decision
751 carbon prices.

752 7. Conclusions

753 This paper studied carbon-aware scheduling of deferrable computing work-
754 loads across data-centre sites as a sustainable-operations problem. We showed
755 that the offline problem, including width limits, deadlines and the emissions
756 of data transfer, is a minimum-cost flow. This yields exact oracles and fast on-
757 line planning. We proved that uncertainty about demand alone forces every
758 online algorithm to a competitive ratio of order $\sqrt{\theta}$, even with perfect carbon
759 forecasts, and that myopic planning does far worse. We proposed CARMA,
760 which augments the planning network with ghost jobs for expected arrivals,
761 and a probabilistic carbon-intensity forecaster with adaptive conformal cali-
762 bration. We proved that CARMA meets every deadline and is θ -competitive
763 under any predictions, optimal under exact predictions and smooth in be-
764 tween. The dual prices of the pooled problem also yield a carbon attribution
765 that is in the core of the pooling game. In 52 out-of-sample weeks of 2024
766 on British regional grid data, CARMA cut emissions relative to carbon-
767 agnostic operation by 36–68% when migration was allowed. That is within
768 6–9% of the clairvoyant optimum and 7.6–9.4% below myopic MPC, and
769 CARMA also reduced late work. Anticipating demand was worth far more
770 than perfect carbon foresight, while risk-averse carbon pricing helped only
771 myopic planners. CARMA degraded gracefully under injected prediction er-
772 rors. Location-based carbon accounting violated coalitional stability in every
773 test week. The dual attribution never did for the hindsight benchmark, and

774 failed only 5 of 728 checks for CARMA’s realised emissions. When jobs can-
775 not migrate, achievable reductions are smaller (9–18%) and forecast accuracy
776 is the binding constraint. Carbon-aware computing is most effective when
777 schedulers plan for the work that has not yet arrived, not only for the grid
778 conditions that lie ahead.

779 **CRediT authorship contribution statement**

780 **Quang-Vinh Dang:** Conceptualization, Methodology, Software, Data
781 curation, Formal analysis, Investigation, Validation, Visualization, Writing –
782 original draft, Writing – review & editing.

783 **Declaration of competing interest**

784 The author declares no known competing financial interests or personal
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790 **Data availability**

791 The carbon-intensity data are publicly available from the NESO Carbon
792 Intensity API [41] under the CC BY 4.0 licence. The data snapshot used in
793 this study, all code, the workload generator with fixed random seeds, and all
794 raw experiment outputs are available from the author’s repository (<https://github.com/vinhqdang/Sustainable-Operations-and-Computers>).
795

796 **Appendix A. Proofs**

797 *Atoms.* Splitting a job j into sub-jobs $j_1, \dots, j_k$ with work w_{j_i} and widths
798 $m_{j_i} = m_j w_{j_i} / w_j$ does not change the optimal value of any planning problem.
799 Given a solution for the merged job, the allocation $x_{j_i s}(t) = x_{j s}(t) w_{j_i} / w_j$
800 satisfies every constraint of the split problem at the same cost. Conversely,
801 summing the sub-job allocations gives a feasible allocation for the merged
802 job, because $\sum_i m_{j_i} = m_j$. Splitting also preserves window-feasibility, since

803 $w_{j_i}/m_{j_i} = w_j/m_j$. We may therefore compare true and predicted job sets
 804 atom by atom.

805 *Appendix A.1. Proof of Theorem 1*

806 Normalise $L = 1$, so $U = \theta$. Site A has one server with cost $p \in [1, \theta]$ in
 807 hour 0 and cost 1 in hour 1. Site B has ample capacity at cost θ in both
 808 hours. Transfer costs are zero. A flexible job ($w = m = 1$, window $\{0, 1\}$) is
 809 released at $t = 0$. The adversary either releases nothing more (scenario 1) or
 810 releases an urgent job ($w = m = 1$, window $\{1\}$) at $t = 1$ (scenario 2). All
 811 costs are known in advance.

812 Consider any deterministic online algorithm that may split work fraction-
 813 ally, and let $x \in [0, 1]$ be the amount of flexible work it runs at $t = 0$. Running
 814 at B in hour 0 costs $\theta \geq p$ and is dominated, so we may assume this work runs
 815 at A . In scenario 1 the algorithm finishes the remaining $1 - x$ at A in hour 1,
 816 so $\text{ALG}_1 = xp + (1 - x)$, while $\text{OPT}_1 = 1$. In scenario 2, hour 1 must process
 817 $2 - x$ units and A can take only one of them, so $\text{ALG}_2 = xp + 1 + (1 - x)\theta$,
 818 while $\text{OPT}_2 = p + 1$. The ratio $r_1(x) = 1 + x(p - 1)$ increases in x , and the
 819 ratio $r_2(x) = (1 + \theta - x(\theta - p))/(p + 1)$ decreases in x . Hence $\min_x \max\{r_1, r_2\}$
 820 is attained where they are equal, at $x^* = (\theta - p)/(p^2 - p - 1 + \theta) \in [0, 1]$,
 821 and its value is

$$1 + \frac{(p - 1)(\theta - p)}{p^2 - p - 1 + \theta}.$$

822 The adversary chooses p to maximise this value, which gives $R^*(\theta)$. The
 823 choice $p = \sqrt{\theta}$ gives the stated closed-form lower bound. For a randomised
 824 algorithm, the expected cost in each scenario is affine in x , so it equals the
 825 cost of the deterministic fractional algorithm that plays $\mathbb{E}[x]$, and the bound
 826 applies against an oblivious adversary. Myopic MPC sees only the flexible
 827 job at $t = 0$ and defers it entirely to the cheaper hour 1 ($x = 0$). With
 828 $p \rightarrow 1^+$, its ratio in scenario 2 tends to $(1 + \theta)/2$. $\square$

829 *Appendix A.2. Proof of Theorem 2*

830 *Robustness.* We first show by induction that every pending real job j is
 831 window-feasible at the start of every hour: its remaining work satisfies $r_j \leq$
 832 $m_j(d_j - t)$. This holds at release by (A1). Suppose it holds at hour t ,
 833 and suppose the plan of CARMA leaves some real work $u_j > 0$ unfinished.
 834 Because $r_j \leq m_j(d_j - t)$, some hour $\tau \in [t, d_j)$ carries less than m_j units of
 835 job j in the plan. Moving a small amount of u_j to the backstop in hour τ
 836 keeps the plan feasible, because the backstop has unlimited capacity, and it

837 changes the cost by a negative amount, because the backstop costs at most
838 $U < \Pi$. This contradicts optimality, so the plan completes all real work.
839 After the first hour is executed, the rest of the plan is a feasible completion,
840 so $r'_j \leq m_j(d_j - t - 1)$ and the induction step holds. No real job is ever late.
841 Every executed server-hour costs at most U , so $\text{ALG} \leq UW$, where W is
842 the total work. Every server-hour in the oracle's solution costs at least L ,
843 and slack costs $\Pi > U \geq L$, so $\text{OPT} \geq LW$. Hence $\text{ALG} \leq \theta \text{OPT}$.

844 *Telescoping.* Let σ_t be the state at the start of hour t : the remaining work
845 of all real jobs released by t . Let $V_t(\sigma, G; c)$ be the optimal value of (2)–(5)
846 over hours $t, \dots, T-1$ for the pending jobs σ and the future jobs G under
847 costs c . Write $\bar{V}_t = V_t(\sigma_t, F_t; c)$, so that $\bar{V}_0 = \text{OPT}$ and $\bar{V}_T = 0$. For a feasible
848 hour- t allocation x with cost $g_t(x)$ and successor state $\sigma^+(x)$, define

$$Q_t(x) = g_t(x) + V_{t+1}(\sigma^+(x), F_t; c), \quad \hat{Q}_t(x) = g_t(x) + V_{t+1}(\sigma^+(x), \hat{F}_t; \hat{c}_t).$$

849 By dynamic programming over a deterministic future, $\min_x Q_t(x) = \bar{V}_t$, and
850 $Q_t(x_t) = g_t(x_t) + \bar{V}_{t+1}$ for the executed action x_t . Summing over t gives

$$\text{ALG} - \text{OPT} = \sum_{t=0}^{T-1} \delta_t, \quad \delta_t = Q_t(x_t) - \min_x Q_t(x) \geq 0.$$

851 Ghost jobs are released at $t+1$ or later, the planning window reaches T
852 by (A2), and hour- t costs are observed. The planning problem solved by
853 CARMA at hour t is therefore exactly $\min_x \hat{Q}_t(x)$, and x_t is a minimiser.

854 *One-step loss.* Let $x^* \in \arg \min Q_t$. Since $\hat{Q}_t(x_t) \leq \hat{Q}_t(x^*)$, we have $\delta_t \leq$
855 $\Delta(x_t) - \Delta(x^*)$, where $\Delta = Q_t - \hat{Q}_t$. Write $\Delta = D_1 + D_2$ with

$$\begin{aligned} D_1(x) &= V_{t+1}(\sigma^+(x), F_t; c) - V_{t+1}(\sigma^+(x), \hat{F}_t; c), \\ D_2(x) &= V_{t+1}(\sigma^+(x), \hat{F}_t; c) - V_{t+1}(\sigma^+(x), \hat{F}_t; \hat{c}_t). \end{aligned}$$

856 *Cost error.* An optimal solution under $\hat{c}_t$ is feasible under c , and its cost
857 changes by at most ε_t per unit of flow on future-hour arcs. The same holds
858 with the roles of c and $\hat{c}_t$ exchanged. Hence $|D_2(x)| \leq \varepsilon_t W_t$ for all x , and
859 $D_2(x_t) - D_2(x^*) \leq 2\varepsilon_t W_t$.

860 *Demand error.* Under (A1), the penalty attached to a window-feasible
861 job does not affect any optimal value, provided it is at least U . If a solution
862 leaves slack on such a job, the argument used for robustness moves that slack

863 to the backstop at no greater cost. True future jobs (penalty Π) and ghost
 864 jobs (penalty Π_g) can therefore be treated as identical atoms whenever they
 865 coincide. Let a be a window-feasible atom and G any set of future jobs. For
 866 every state σ ,

$$Lw_a \leq V(\sigma, G \cup \{a\}) - V(\sigma, G) \leq Uw_a.$$

867 For the upper bound, add a to an optimal solution for G by running it on the
 868 backstop at rate $w_a/(d_a - a_a) \leq m_a$ in every hour of its window. For the lower
 869 bound, delete a 's flow and slack from an optimal solution for $G \cup \{a\}$; each
 870 deleted unit cost at least $\min\{L, \Pi_a\} = L$, since $\Pi_a \geq U \geq L$. Now transform
 871 $\hat{F}_t$ into F_t by removing $\hat{F}_t \setminus F_t$ and adding $F_t \setminus \hat{F}_t$, one atom at a time. Then
 872 $D_1(x)$ is a sum of increments, each lying in $[Lw_a, Uw_a]$ (with a minus sign
 873 for removals) for every x . Therefore $D_1(x_t) - D_1(x^*) \leq (U - L) \sum_a w_a =$
 874 $(U - L) d_t$.

875 Combining the two bounds gives $\delta_t \leq (U - L)d_t + 2\varepsilon_t W_t$. Summing over t
 876 and taking the minimum with the robustness bound proves (9). If $d_t = \varepsilon_t = 0$
 877 for all t , then $\delta_t = 0$ and $\text{ALG} = \text{OPT}$. $\square$

878 *Appendix A.3. Proof of Theorem 3*

879 Write the problem of coalition C as $c(C) = \min\{c^\top z : A_+=z = b_+(C), A_+z \leq$
 880 $b_+(C), z \geq 0\}$, where $z = (x, u)$. Jobs submitted outside C have zero work
 881 and zero width, and sites outside C have zero capacity. This forces the
 882 corresponding flows to zero, so the problem coincides with the stand-alone
 883 problem of C . The right-hand side is additive over sites: $b(C) = \sum_{o \in C} b^o$,
 884 where b^o collects the work and widths of the jobs submitted at o and the
 885 capacity of site o . The matrix A and the cost c do not depend on C . The
 886 dual problem $\max\{b(C)^\top y : A^\top y \leq c, y_- \leq 0\}$ therefore has a feasible set
 887 that is the same for all coalitions.

888 Let $y^* = (\lambda, \nu, \mu)$ be an optimal dual solution for $\mathcal{S}$. The slack variables
 889 make every primal problem feasible, and costs are non-negative, so strong
 890 duality gives $c(\mathcal{S}) = b(\mathcal{S})^\top y^* = \sum_o (b^o)^\top y^* = \sum_o \phi_o$, which is budget balance.
 891 For any coalition C , y^* is dual-feasible for C , so weak duality gives $c(C) \geq$
 892 $b(C)^\top y^* = \sum_{o \in C} \phi_o$, which is the core condition [27]. Expanding $(b^o)^\top y^*$
 893 gives (10). By Proposition 1 the problem is a min-cost flow. Its optimal
 894 node potentials, together with the reduced costs they induce on saturated
 895 capacity arcs, give an optimal dual solution, so network simplex returns y^*
 896 together with the optimal flow. $\square$

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